https://fumeng-yang.github.io/CMSC471/

CMSC471 Home HTW Syllabus Crew Policies

Introduction to Information Visualization

Time: MW 2:00pm - 3:15pm | **Location:** CSI 3117 | **Semester:** Spring 2025

Last update: 01/22/2025 21:56

Description—Datasets are becoming increasingly large and complex, requiring intuitive ways to explore and interpret them quickly and efficiently. In this case, a picture is worth a thousand words: visualizations enable us to transform data into images that are easier to understand and reason about, compared to raw numbers and raw text. Visualizations are critical tools in externalizing and organizing our knowledge and insights, whether to explore collected datasets to improve our understanding of the physical world, to assess and debug analysis/experimental workflows, or to present new and interesting results to diverse audiences. In this course we will study techniques and algorithms for creating effective visualizations based on principles from graphic design, perceptual psychology, and cognitive science. Students will learn how to design and build interactive visualizations for the web, using the D3.js (Data-Driven Documents) framework.

Course objectives—By the end of this course, students can

- understand key visualization techniques and theory.
- implement interactive data visualizations using D3.js.
- design, evaluate, and critique visualization designs.
- develop a substantial visualization project.

Note: We reserve the right to add, remove, or alter the syllabus as needed throughout the semester. Major changes will be notified before the release.

Useful links: Canvas Piazza

Teaching assistants

Zhongzheng Xu

"jong-jung"

Taehyun Yang

"tay-hyun"